

Size structure of *Sperata aor* (Gangetic Tengra) from the Ghaghara river at Ayodhya, India

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Abstract

The size of *Sperata aor* varied from 9.1 to 119.0 cm in November 2017 to October 2018 from the Ghaghara river at Ayodhya, Uttar Pradesh. The maximum exploitation was recorded in 59.1 to 69.0 cm size group with 19.70% but minimum exploitation was estimated in 109.1-119.0 cm size group (1.14%). The exploitation was observed 3.41 in 9.1-19.0 cm, 6.44% in 19.1-29.0 cm, 10.61% in 29.1-39.0 cm, 13.26% in 39.1-49.0 cm, 17.80% in 49.1-59.0 cm, 14.39% in 69.0-79.1 cm, 4.54% in 89.0-99.0 cm and 2.65% in 99.0-109.0 cm size groups. Overall, middle size group shared maximum exploitation compared to lower size group and higher size group. Results also indicated that the exploitation was not systematic in middle size group to higher size group.

Keywords: Size structure, Exploitation pattern, Ghaghara river, *Sperata aor*

Introduction

The Gangetic Tengra, *Sperata aor* very common fish species from the Ganga river system (Dwivedi *et al.* 2016a, Tripathi *et al.* 2017a) and choose riverine habitat (Chondar 1999). It constitutes the chief component of the fishery in the middle and lower stretches of the river (Talwar & Jhingran 1991, Tripathi *et al.* 2017b). It is a very popular food fish in the Ganga basin due to its good taste and high nutritional value (Mayank *et al.* 2018). *S. aor* is more abundant than *S. seenghala* in the nature. Its natural distribution is in the freshwater of India, Nepal, Bangladesh, Pakistan, and Burma (Chondar 1999). The objective of the present study was to give size structure and exploitation of the *S. aor* from the Ghaghara river at Ayodhya, Uttar Pradesh.

Material and methods

The random fish samples were collected during November 2017 to October 2018 from the Ghaghara river at Ayodhya, Uttar Pradesh. Fishes were captured using a variety of methods including drag netting (mahajal, chaundi, darwari), cast netting, gill netting and hook and line. Size of fishes was measured by

simple measuring scale. Total length (TL) of fish was measured from tip of the snout to the largest rays of caudal fin. Collected data were classified at 10 cm intervals. The number of samples calculated according to size group then converted into percentage.

Result and discussion

The size of *Sperata aor* varied from 9.1 to 119.0 cm from the Ghaghara river at Ayodhya, Uttar Pradesh (Table 1). The maximum exploitation was recorded in 59.1 to 69.0 cm size group with 19.70% but minimum exploitation was estimated in 109.1-119.0 cm size group (1.14%). The exploitation was observed 3.41 in 9.1-19.0 cm, 6.44% in 19.1-29.0 cm, 10.61% in 29.1-39.0 cm, 13.26% in 39.1-49.0 cm, 17.80% in 49.1-59.0 cm, 14.39% in 69.0-79.1 cm, 4.54% in 89.0-99.0 cm and 2.65% in 99.0-109.0 cm size groups. Overall, middle size group shared maximum exploitation compared to lower size group and higher size group. Results also indicated that the exploitation was not systematic in middle size group to higher size group. If higher size group exploitation is not systematic then it is an alarming sign for future population of *S. aor*. In collective, middle size group

of fishes in case of riverine/natural stocks are more exploited than other size groups (Nikolskii 1980, Pathak *et al.* 2011, 2015, 2015ba, Tripathi *et al.* 2015). Imran *et al.* (2015), Mayank & Dwivedi (2015b, 2016), Dwivedi *et al.* (2016b) stated that the middle size groups of any commercially important fish species have more attractive to the fishermen and consumers. Tiwari *et al.* (2016, 2017) and Dwivedi *et al.* (2018) also reported similar findings in the Ganga river. The catfishes are the backbone of the Ganga river fishery (Mayank & Dwivedi 2015b). The water discharge of the river also interferes in the production of the culture pond of the surrounding river bank (Jha *et al.* 2015, Dwivedi *et al.* 2018). Exploitation is an economic activity governed by social needs and pressures (Dwivedi *et al.* 2014, Mayank & Dwivedi 2016b). The over exploitation, water pollution and targeted fishing is the main problem of the riverine

fishery stock (Jatan *et al.* 2014, Tiwari *et al.* 2016, Dwivedi *et al.* 2016c, 2017, Muthmainnah and Gaffar 2017, Mayank *et al.* 2018).

Table 1 Size structure of *Sperata aor* from the Ghaghara river at Ayodhya

SN	Size groups (cm)	No. of samples	Percentage
1	9.1-19.0	09	3.41
2	19.1-29.0	17	6.44
3	29.1-39.0	28	10.61
4	39.1-49.0	35	13.26
5	49.1-59.0	47	17.80
6	59.1-69.0	52	19.70
7	69.1-79.0	38	14.39
8	79.1-89.0	16	6.06
9	89.1-99.0	12	4.54
10	99.1-109.0	7	2.65
11	109.1-119.0	03	1.14

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